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CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1 Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling. Starting from a fully connected autoencoder, the experiment progresses to a Convolutional Autoencoder (CAE), introduces controlled image corruption for a denoising autoencoder, and finally implements a Variational Autoencoder (VAE). The learned latent representations are visualized, reconstruction quality is compared using MSE, MAE and SSIM, and the generative capability of the VAE is explored through sampling and latent interpolation. All four models were trained under a controlled protocol on the same MNIST subset.

2 Learning Outcomes

- Explain the encoder–latent–decoder structure of an autoencoder.
- Prepare image data for reconstruction-based learning.
- Implement a fully connected autoencoder and a Convolutional Autoencoder.
- Introduce controlled noise and build a denoising autoencoder.
- Evaluate reconstruction using MSE, MAE and SSIM.
- Visualize and interpret latent representations.
- Implement the VAE reparameterization trick and interpret reconstruction/KL losses.
- Generate new images by sampling from a learned latent space.
- Compare deterministic and variational latent representations.

3 Dataset and Experimental Setup

The MNIST Handwritten Digit Dataset was used, loaded directly via `tensorflow.keras.datasets.mnist`. Each image has spatial dimensions $28 \times 28 \times 1$. As recommended for the laboratory execution, a fixed subset of **10,000 training images** and **2,000 test images** was randomly sampled (seed = 42) from the full 60,000/10,000 MNIST split. Labels were used only for latent-space visualization and were never supplied as reconstruction targets. Pixel values were normalized from $[0, 255]$ to $[0, 1]$.

Property	**Value**
Full training samples	60,000
Full test samples	10,000
Selected training samples	10,000
Selected test samples	2,000
Image shape	$28 \times 28 \times 1$
Flattened vector length	784
Normalized pixel range	[0.00, 1.00]
Number of classes	10
Device used	GPU (T4)



Figure 1: Sample MNIST training images from the selected subset.

4 Expected Input and Output

For the fully connected autoencoder, each image is flattened to $x \in \mathbb{R}^{784}$. For the convolutional, denoising and variational models, the spatial structure is retained as $x \in \mathbb{R}^{28\times28\times1}$. In every case the output is a reconstructed image $\hat{x} \in \mathbb{R}^{28\times28\times1}$ (or its flattened equivalent). For the denoising model the output additionally includes the noisy input, and for the VAE the output additionally includes newly generated samples obtained by decoding vectors drawn from the latent prior.

5 Overall Experimental Pipeline

MNIST → Normalization → Encoder → Latent Representation → Decoder → Reconstructed Image → Evaluation

The experiment consists of four related studies executed in sequence: (1) Fully Connected Autoencoder, (2) Convolutional Autoencoder, (3) Denoising Convolutional Autoencoder, and (4) Variational Autoencoder, followed by a latent-dimension study and eight additional exercises.

6 Exploring Autoencoders

An autoencoder learns to reproduce its input at its output. The encoder maps the input to a lower-dimensional latent representation $z = f_\theta(x)$, and the decoder reconstructs the input $\hat{x} = g_\phi(z)$. The training objective minimizes reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N}\sum_{i=1}^{N}\|x_i - \hat{x}_i\|^2.$$

The latent vector z is a compressed representation of the input image and forms the bottleneck through which all information relevant to reconstruction must pass.

7 Fully Connected Autoencoder

The FC-AE follows the architecture 784 → 128 → 32 → 16(latent) → 32 → 128 → 784, with ReLU activations in the hidden layers and a sigmoid output activation. It was trained with the Adam optimizer (learning rate 10^{-3}), batch size 128, for 20 epochs, using Binary Cross-Entropy loss.

Parameter	**Value**
Latent dimension	16
Trainable parameters	211,040
Optimizer	Adam ($\eta = 10^{-3}$)
Batch size	128
Epochs	20
Training time	13.61 s
Final training loss (BCE)	0.12984
Final validation loss (BCE)	0.12939

8 Required Analysis of the Basic Autoencoder

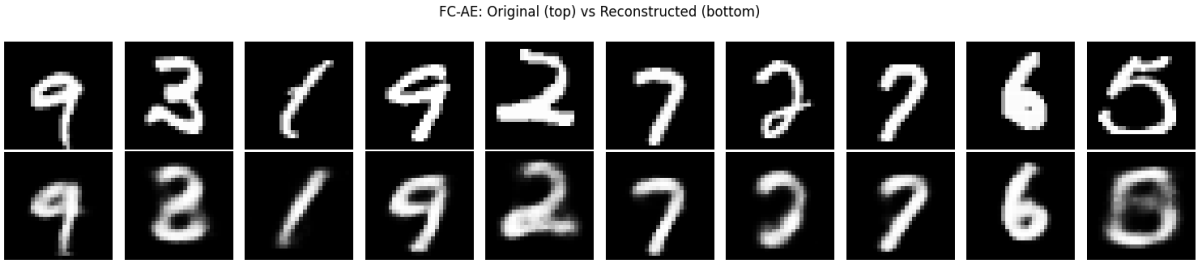


Figure 2: FC-AE: original (top row) versus reconstructed (bottom row) test images.

The overall digit shape is preserved for every sample, but the reconstructions are visibly blurred and fine strokes are smoothed out. Thin, cursive-style digits (e.g. the 1 and 2) lose sharp edges, and the last sample (a genuine 5) is reconstructed with an ambiguous loop that resembles an 8 – a difficult case where flattening the image discards the spatial cues that would otherwise disambiguate the stroke. Easy, well-separated digits such as 7 and 9 reconstruct considerably more cleanly than digits with overlapping or curved strokes.

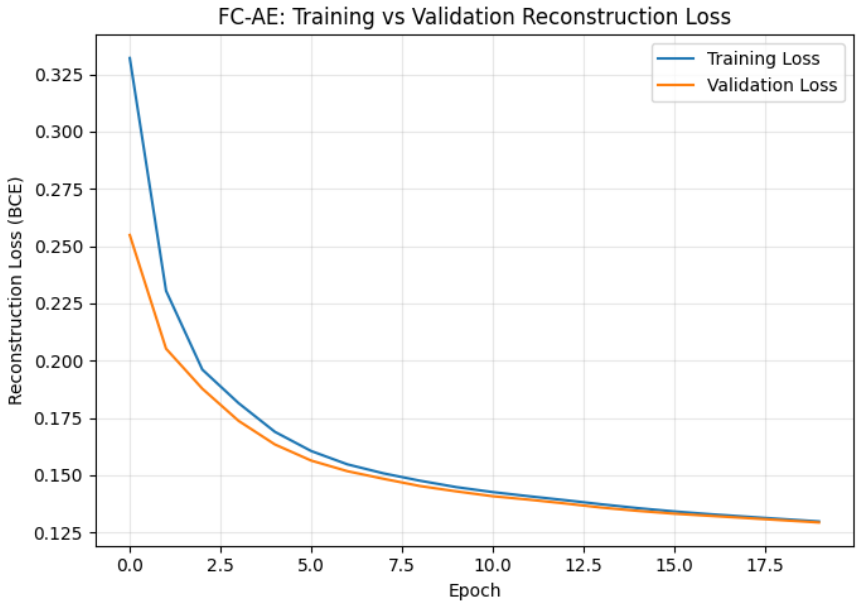


Figure 3: FC-AE training and validation reconstruction loss over 20 epochs.

Both curves decrease smoothly and converge to nearby values (final training loss 0.12984, final validation loss 0.12939, a gap of only 0.00045), indicating stable convergence with no sign of overfitting on this subset.

9 Reconstruction Metrics

Metric	Value
Test MSE	0.022270
Test MAE	0.059194
Mean SSIM	0.745986

10 Convolutional Autoencoder

The CAE follows `Conv2D(32)`→`MaxPool`→`Conv2D(64)`→`MaxPool`→`Conv2D(64)`→`UpSampling` →`Conv2D(32)`→`UpSampling`→`C` trained under the identical protocol as the FC-AE (Adam, $\eta = 10^{-3}$, batch size 128, 20 epochs, BCE loss) for a controlled comparison.

Parameter	Value
Trainable parameters	74,497
Training time	24.31 s
Final training loss (BCE)	0.06843
Final validation loss (BCE)	0.06786
Test MSE	0.002419
Test MAE	0.014711
Mean SSIM	0.975349

11 Comparing Fully Connected and Convolutional Autoencoders

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.02227	0.05919	0.74599	211,040
Convolutional AE	0.00242	0.01471	0.97535	74,497

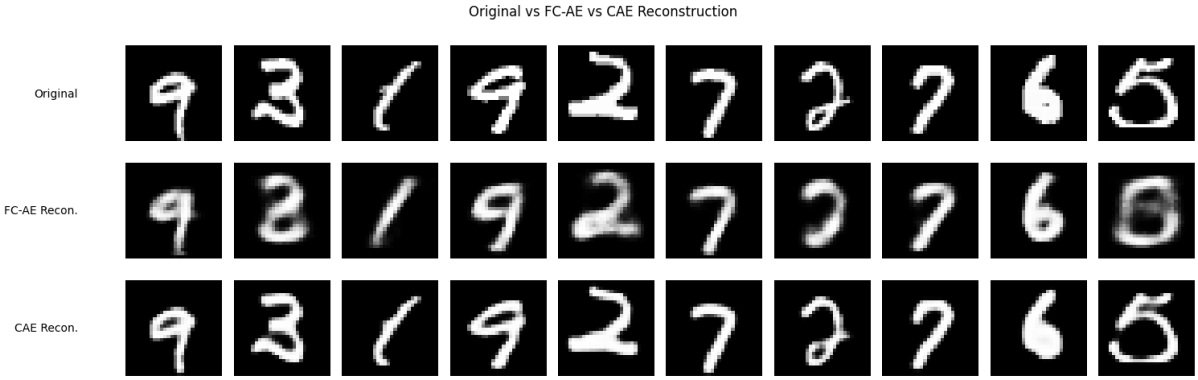


Figure 4: Original versus FC-AE reconstruction versus CAE reconstruction on the same test samples.

The CAE achieves roughly a 9-fold lower MSE and a 15-percentage-point higher SSIM than the FC-AE while using *fewer* parameters (74,497 versus 211,040). By preserving spatial locality through convolution and pooling instead of flattening the image into a vector, the CAE retains stroke geometry and edges far more faithfully – visible directly in Figure 4, where the CAE reconstructions are sharper and the ambiguous 5→8 confusion seen in the FC-AE disappears entirely.

12 Denoising Autoencoder

The denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image: $\tilde{x} \rightarrow$ Encoder $\rightarrow z \rightarrow$ Decoder $\rightarrow \hat{x}$, with the clean image always used as the target. The same convolutional architecture as the CAE was reused for the encoder–decoder.

13 Adding Image Noise

Two corruption types were implemented. Gaussian noise $\tilde{x} = x + n,\ n \sim \mathcal{N}(0, \sigma^2)$ was applied at $\sigma \in \{0.1, 0.2, 0.3\}$ with pixel clipping to $[0, 1]$. Salt-and-pepper noise was applied at corruption levels $p \in \{0.05, 0.10, 0.20\}$, randomly replacing pixels with 0 or 1. The main denoising model was trained using Gaussian noise at $\sigma = 0.2$.

Parameter	Value
Training noise type/level	Gaussian, $\sigma = 0.2$
Trainable parameters	74,497
Training time	20.41 s
Final training loss (BCE)	0.07548
Final validation loss (BCE)	0.07505
Test MSE	0.004468
Test MAE	0.021070
Mean SSIM	0.948457

14 Required Denoising Plots

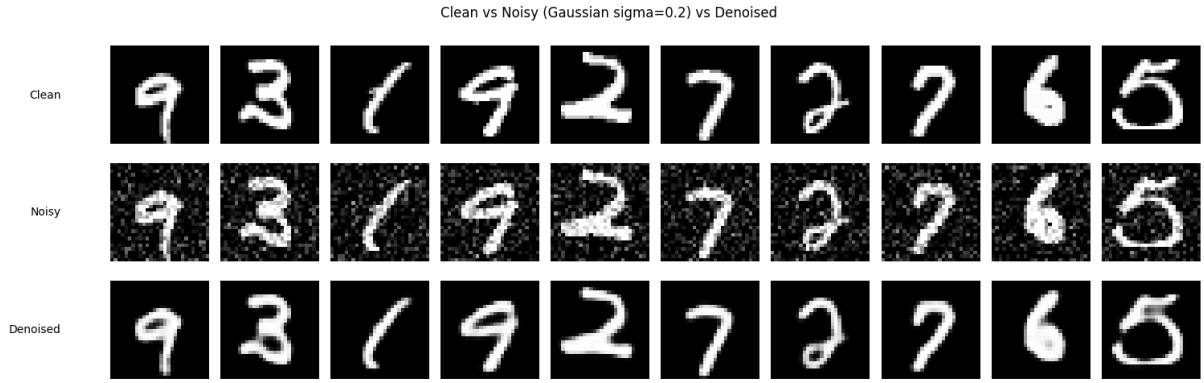


Figure 5: Clean versus noisy (Gaussian $\sigma = 0.2$) versus denoised test images.

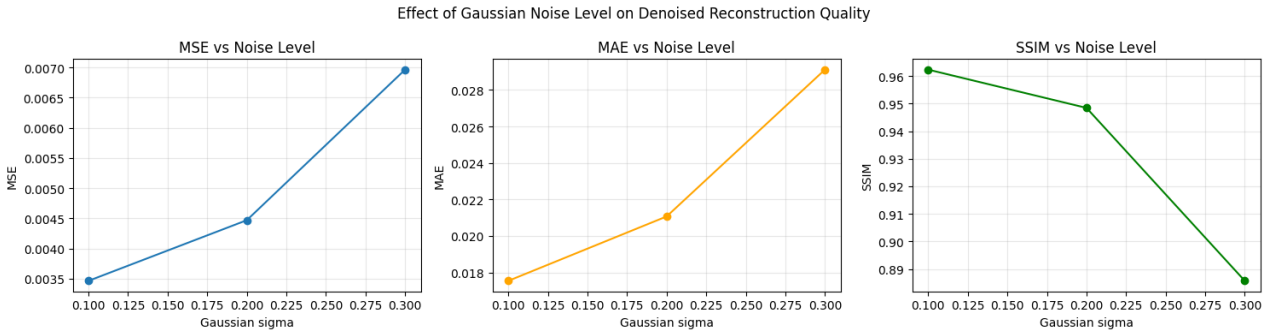


Figure 6: Effect of Gaussian noise level on denoised reconstruction quality (MSE, MAE, SSIM).

Noise Level (σ)	MSE	MAE	SSIM
0.1	0.003462	0.017543	0.962333
0.2	0.004468	0.021070	0.948457
0.3	0.006966	0.029112	0.885719

Salt-Pepper Level (p)	MSE	MAE	SSIM
0.05	0.004757	0.021012	0.934433
0.10	0.006981	0.027018	0.888367
0.20	0.013783	0.043147	0.767158

As the Gaussian noise level increases from $\sigma = 0.1$ to $\sigma = 0.3$, MSE and MAE roughly double while SSIM drops from 0.962 to 0.886. The model, trained only at $\sigma = 0.2$, still generalizes reasonably to nearby noise levels but degrades faster outside its training level, and is evaluated here on salt-and-pepper noise it never saw during training – its SSIM still declines with increasing corruption ($p = 0.05 \rightarrow 0.20$: SSIM $0.934 \rightarrow 0.767$), confirming that the model recovers major digit structure under both noise types but loses fine detail as corruption grows more severe.

15 Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector $z = f(x)$. A VAE instead learns a distribution $q_\phi(z|x) = \mathcal{N}(\mu(x), \mathrm{diag}(\sigma^2(x)))$. The encoder outputs $\mu, \log \sigma^2$, and a latent vector is drawn using the reparameterization trick $z = \mu + \sigma \odot \epsilon,\ \epsilon \sim \mathcal{N}(0, I)$, which keeps the sampling step differentiable with respect to the encoder parameters.

16 VAE Loss Function

The VAE objective combines a Binary Cross-Entropy reconstruction term with the closed-form KL divergence against the standard normal prior $p(z) = \mathcal{N}(0, I)$:

$$D_{KL}\big(q_\phi(z|x) \,\|\, p(z)\big) = -\frac{1}{2}\sum_j \left(1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2\right), \qquad \mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}.$$

17 VAE Implementation

A compact convolutional encoder/decoder was used with a 2-dimensional latent space (chosen for direct visualization). The encoder produces $\mu, \log \sigma^2 \in \mathbb{R}^2$; the decoder maps $z \in \mathbb{R}^2$ back to a $28 \times 28 \times 1$ image via dense and transposed-convolution layers. The model (encoder + decoder) has 284,933 trainable parameters and was trained for 30 epochs (Adam, $\eta = 10^{-3}$), taking 32.42 s.

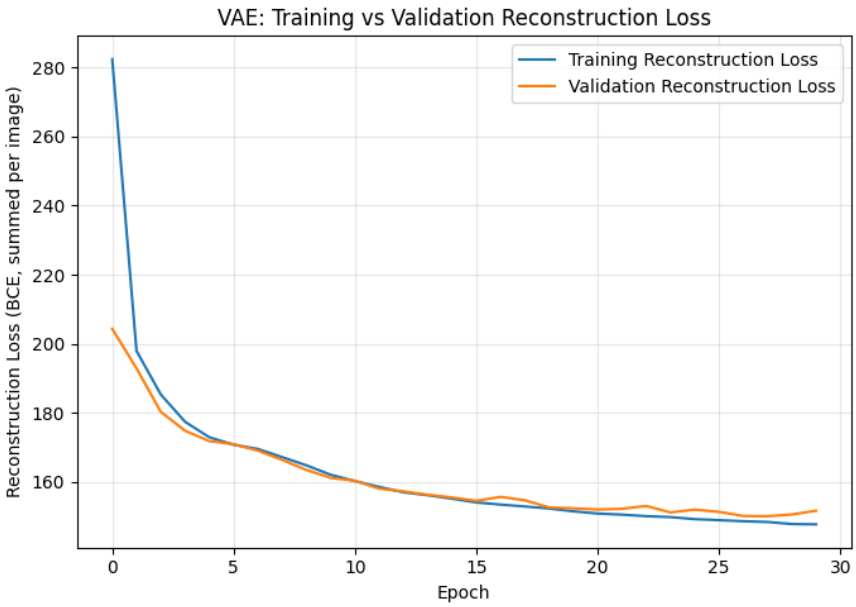


Figure 7: VAE training versus validation reconstruction loss (summed BCE per image).

Both curves fall sharply in the first few epochs and then flatten, with validation loss tracking training loss closely throughout (final training reconstruction loss 147.67, final validation reconstruction loss 151.61), indicating stable convergence without significant overfitting.

VAE Metric	Value
Reconstruction Loss (val)	151.6052
KL Loss (val)	6.2382
Total Loss (val)	157.8434
Test MSE	0.044460
Test MAE	0.101560
Mean SSIM	0.516786

The VAE's reconstruction metrics (MSE, SSIM) are visibly weaker than the deterministic CAE's, which is expected: compressing all digit variation into only 2 latent dimensions, combined with the KL penalty that regularizes the latent space toward $\mathcal{N}(0, I)$, trades reconstruction sharpness for a smooth, generative latent space.

18 Exploring the VAE Latent Space

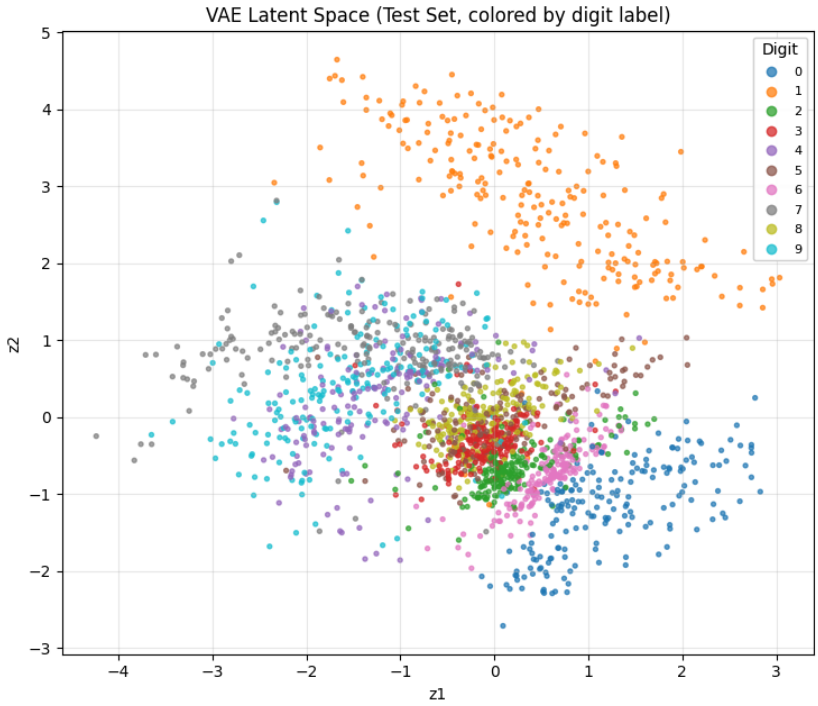


Figure 8: VAE latent space (z_1, z_2) of the test set, colored by digit label (labels used only for visualization).

The encoded test points (z_{mean} range: $x \in [-4.23, 3.04]$, $y \in [-2.71, 4.65]$) form recognizable clusters: digit 1 occupies a distinct region at high z_2, digit 0 separates toward positive z_1/negative z_2, and 3, 4, 5, 8 overlap heavily near the origin because these digits share similar curved-stroke structure that a 2-D bottleneck cannot fully disentangle. The overlap in the dense central region is consistent with the VAE's comparatively lower SSIM, since visually similar digits are mapped to nearby latent codes and become harder for the decoder to reconstruct unambiguously.

19 Generating New Images with the VAE

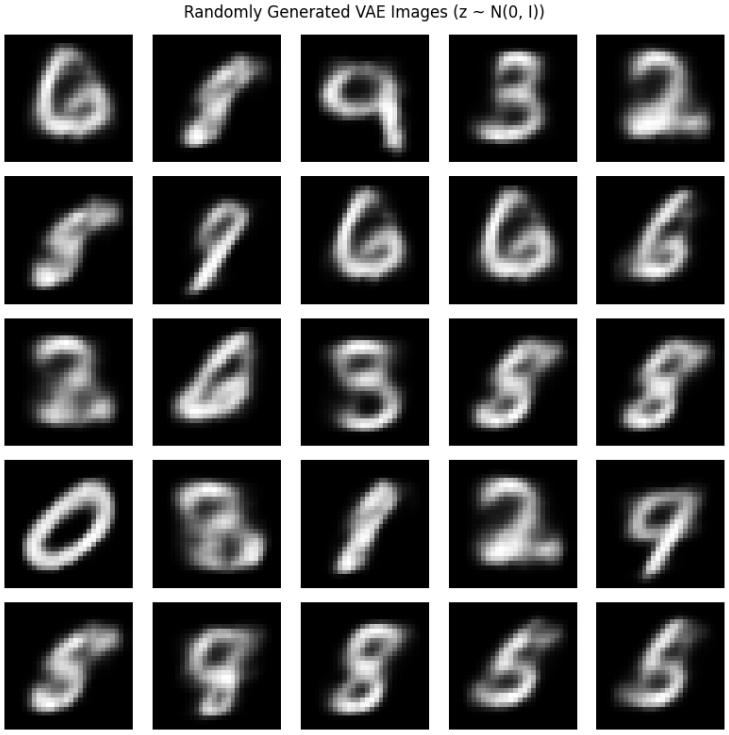


Figure 9: 25 images generated by sampling $z \sim \mathcal{N}(0, I)$ and decoding.

Many of the generated samples are clearly recognizable as digits (6, 9, 3, 2, 0 are legible), confirming that the decoder has learned a meaningful generative mapping from the prior. There is visible variation in stroke style and digit identity across the grid, but several samples are blurred or ambiguous between two digits (e.g. shapes between 5 and 6, or 3 and 8) – a direct consequence of the overlapping latent regions seen in Figure 8 and the limited capacity of a 2-D latent space.

20 Latent Space Interpolation

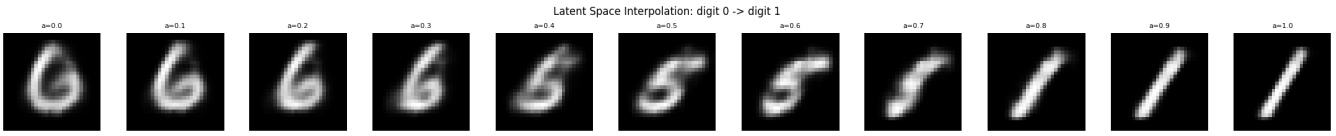


Figure 10: Decoded images along the linear interpolation $z(\alpha) = (1 - \alpha)z_A + \alpha z_B$ between a digit-0 and a digit-1 latent code, $\alpha = 0, 0.1, \ldots, 1.0$.

The decoded image changes smoothly and continuously as α increases: the initial frames resemble a 6-like loop (reflecting the chosen test sample's z-position rather than a clean 0), passing through a 5-like intermediate shape before straightening into a 1 by $\alpha = 0.8$–1.0. No abrupt jumps or discontinuities appear, demonstrating that the VAE's latent space is locally continuous and semantically organized.

21 Reconstruction Comparison Across Models

Model	**MSE**	**MAE**	**SSIM**	**Parameters**
Fully Connected AE	0.02227	0.05919	0.74599	211,040
Convolutional AE	0.00242	0.01471	0.97535	74,497
Denoising CAE	0.00447	0.02107	0.94846	74,497

VAE Metric	Value
Reconstruction Loss	151.6052
KL Loss	6.2382
Total Loss	157.8434
Test MSE	0.044460
Test MAE	0.101560
Mean SSIM	0.516786

22 Reconstruction Error Distribution

Per-image reconstruction error $e_i = \frac{1}{D}\sum_j (x_{ij} - \hat{x}_{ij})^2$ was computed on the CAE test-set reconstructions.

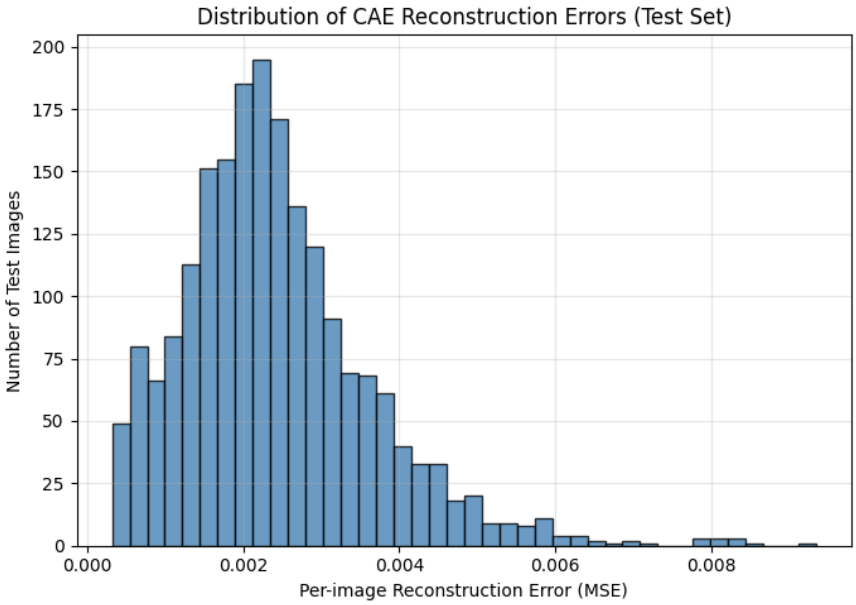


Figure 11: Distribution of per-image CAE reconstruction errors on the test set.

Statistic	Value
Minimum error	0.000313
Maximum error	0.009349
Mean error	0.002419
Median error	0.002256
90th percentile	0.003957
95th percentile	0.004605
99th percentile	0.006195

The distribution is unimodal and right-skewed, concentrated between roughly 0.0005 and 0.004 with a long thin tail extending to about 0.0093. The bulk of test images are reconstructed very accurately (median 0.00226), while a small minority of harder, atypically-drawn samples drive the tail.

23 Required High-Error Image Analysis

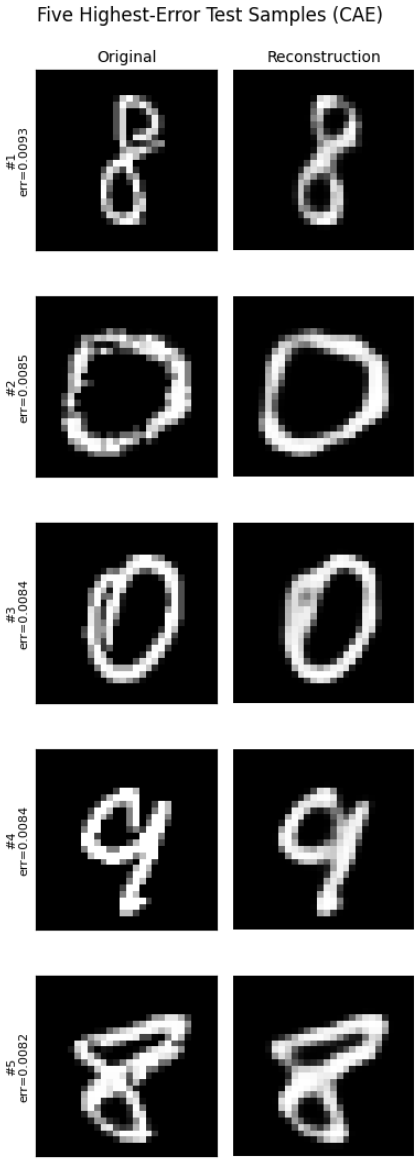


Figure 12: Five highest-error CAE test samples: original (left) versus reconstruction (right).

Rank	Test Index	True Digit	MSE
1	589	8	0.009349
2	1317	0	0.008477
3	434	0	0.008397
4	949	9	0.008377
5	152	8	0.008226

Four of the five hardest samples are 0s and 8s, drawn with unusually thick strokes, tight loops or an open/closed ambiguity (e.g. an 8 with a nearly closed lower loop resembling a 0). Such heavy, atypical stroke widths sit further from the bulk of the training distribution, so the encoder's compressed representation loses more of the fine loop geometry, producing visibly thinner or slightly misshapen reconstructions.

24 Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time (s)
FC Autoencoder	0.02227	0.05919	0.74599	211,040	13.61
Conv. Autoencoder	0.00242	0.01471	0.97535	74,497	24.31
Denoising CAE	0.00447	0.02107	0.94846	74,497	20.41
VAE	0.04446	0.10156	0.51679	284,933	32.42

The Convolutional Autoencoder achieved the best pure reconstruction quality among all four models, followed by the Denoising CAE (evaluated on noisy inputs), the FC-AE, and finally the VAE, whose latent space is deliberately regularized for generation rather than optimized purely for reconstruction fidelity.

25 Additional Latent-Dimension Study

The fully connected autoencoder was retrained with latent dimensions 2, 8, 16 and 32 (20 epochs each, identical protocol).

Latent Dim	MSE	SSIM	Parameters	Time (s)
2	0.04718	0.47048	210,130	12.50
8	0.02756	0.69317	210,520	11.67
16	0.02091	0.76230	211,040	11.15
32	0.01837	0.79033	212,080	13.33

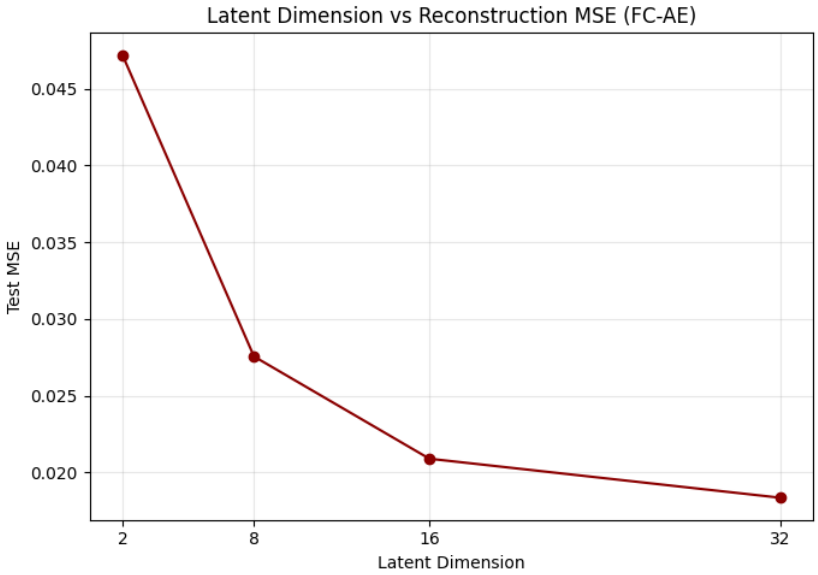


Figure 13: Latent dimension versus test-set reconstruction MSE for the FC-AE.

MSE decreases and SSIM increases monotonically as the latent dimension grows from 2 to 32, but with diminishing returns: the largest single improvement is between dim 2 and dim 8 (MSE 0.047 $\rightarrow$ 0.028), while the gain from 16 to 32 is comparatively small (MSE 0.021 $\rightarrow$ 0.018). This illustrates the classic compression-versus-quality trade-off – a larger bottleneck retains more information and reconstructs better, but the marginal benefit shrinks once the latent space is large enough to capture most of the digit variation.

26 Required Plots

All plots required by the experiment specification (original vs. reconstructed, training/validation loss, FC-AE vs. CAE, clean/noisy/denoised, noise level vs. metrics, VAE latent space, generated samples, latent interpolation, VAE training/validation loss, and the reconstruction-error distribution) are presented in the corresponding sections above (Figures 1–10), each with its own inference discussing what the plot shows, the trend observed, why the trend likely occurs, and the conclusion drawn from it.

27 Required Inferences

- **Effect of latent dimension on reconstruction:** MSE falls and SSIM rises monotonically from dim 2 to dim 32 (Table in Sec. 25), with diminishing gains at larger sizes – larger bottlenecks retain more digit information but returns saturate once most variance is captured.
- **FC vs. convolutional reconstruction:** the CAE outperforms the FC-AE on every metric while using fewer parameters, because convolution preserves 2-D spatial locality that flattening destroys.
- **Effect of Gaussian noise level:** SSIM falls from 0.962 to 0.886 as σ rises from 0.1 to 0.3; larger corruption removes more true signal, so the denoiser has less to work with.
- **Denoising autoencoder's ability to recover structure:** even at the highest tested noise levels the digit outline remains recognizable in Figure 5, showing the model recovers global structure better than fine strokes.
- **VAE latent distribution characteristics:** the encoder produces smoothly varying $\mu, \log \sigma^2$ values that keep the aggregate posterior close to $\mathcal{N}(0, I)$, evidenced by the KL loss stabilizing near 6.2 rather than growing unbounded.
- **Structure of the 2-D VAE latent space:** distinct digits form loosely separated clusters with substantial overlap near the origin (Figure 8), reflecting genuine visual similarity between digits such as 3/5/8.
- **Quality and diversity of generated samples:** most of the 25 generated digits in Figure 9 are recognizable and show real stylistic variation, though a few fall in ambiguous latent regions and look intermediate between two digits.
- **Smoothness of latent interpolation:** Figure 10 shows a continuous, gradual shape change with no discontinuities between the two endpoint digits.
- **Reconstruction error vs. visual quality:** the five highest-error samples (Sec. 16) are visibly atypical (unusually thick or ambiguous strokes), directly linking numerical error to perceptual difficulty.
- **Deterministic vs. probabilistic latent representations:** the CAE's deterministic code achieves far higher SSIM (0.975) than the VAE's probabilistic code (0.517) at comparable capacity, because the VAE trades reconstruction sharpness for a smooth, continuous, generative latent space.

28 Discussion Questions

1. **What is an autoencoder?** A neural network trained to reconstruct its own input through a compressed intermediate (latent) representation, learning efficient encodings without labels.
2. **Purpose of the encoder?** Compresses the input into a lower-dimensional latent vector that retains the information needed to reconstruct the image.
3. **Purpose of the decoder?** Maps the latent vector back to the original input space, reconstructing an approximation of the input image.
4. **What is a latent representation?** A compact, lower-dimensional vector z that summarizes the essential content of the input, produced by the encoder.
5. **Why is the latent representation usually smaller than the input?** To force the network to learn a compressed, non-trivial encoding rather than simply copying the input through.
6. **What happens when the latent dimension is too small?** Reconstruction quality degrades (higher MSE, lower SSIM) because too little information can pass through the bottleneck, as seen at dim=2 (MSE 0.047).
7. **What happens when the latent dimension is increased?** Reconstruction quality improves with diminishing returns, since more capacity is available to retain fine detail, up to a saturation point.
8. **Why can a CAE preserve spatial information more effectively?** Convolutional filters operate locally and share weights across spatial positions, preserving 2-D structure that a fully-connected flatten operation destroys.
9. **Differentiate an autoencoder and a denoising autoencoder.** A plain autoencoder reconstructs its own clean input; a denoising autoencoder receives a corrupted input but is trained against the clean target, forcing it to learn noise-robust features.
10. **Why is the clean image used as the target in denoising?** So the network learns to recover the true underlying signal rather than simply copying the corrupted input.
11. **What happens when the noise level is increased?** Reconstruction quality worsens monotonically (SSIM fell from 0.962 to 0.886 as σ rose from 0.1 to 0.3), since more true signal is destroyed.
12. **Why are MSE, MAE and SSIM useful for image reconstruction?** MSE/MAE quantify pixel-level error magnitude, while SSIM captures perceptual/structural similarity, so together they give both a numerical and a visually meaningful measure of quality.
13. **What is a Variational Autoencoder?** An autoencoder that learns a probability distribution over the latent space instead of a single deterministic vector, enabling generation of new samples.
14. **How does a VAE differ from a conventional autoencoder?** It encodes inputs as a distribution $(\mu, \log \sigma^2)$ and samples from it, and its loss adds a KL-divergence regularizer to the prior.
15. **Why does a VAE learn a distribution instead of a single latent vector?** A distribution makes the latent space continuous and generative, allowing new, unseen points to decode into plausible images.
16. **What is the reparameterization trick?** Writing $z = \mu + \sigma \odot \epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$ so gradients can flow through μ, σ despite the stochastic sampling step.

17. **Why is ϵ sampled from a standard normal distribution?** It provides the external source of randomness needed for sampling while keeping the transformation to z differentiable in μ, σ.
18. **Role of KL divergence in the VAE loss?** Regularizes the learned posterior $q_\phi(z|x)$ toward the prior $\mathcal{N}(0, I)$, keeping the latent space smooth, continuous and suitable for sampling.
19. **What is the prior distribution used in the VAE?** The standard normal distribution $p(z) = \mathcal{N}(0, I)$.
20. **Why is a 2-D latent space useful for visualization?** It can be plotted directly on a 2-D scatter plot, allowing the cluster structure and digit relationships to be inspected visually, as in Figure 8.
21. **What does latent-space interpolation demonstrate?** That the latent space is continuous and semantically organized, since intermediate points decode into plausible in-between images (Figure 10).
22. **Why can a VAE generate new images?** Because its prior $\mathcal{N}(0, I)$ covers the entire latent space with a known, simple distribution, any sampled point can be decoded into a plausible image.
23. **Why might some VAE-generated images be unclear?** They may fall in overlapping latent regions shared by multiple digit classes (Figure 8), producing decoded images that blend features of two digits.
24. **Difference between reconstruction and generation?** Reconstruction decodes the latent code of a real, encoded input; generation decodes an arbitrary latent code sampled from the prior, with no corresponding real input.
25. **Why should test images remain unseen during training?** To obtain an unbiased estimate of how well the model generalizes to new data rather than measuring memorization of the training set.

29 Additional Exercises

Exercise 1 – CAE with latent channels 4/8/16/32 (10 epochs each): SSIM rose steadily with capacity (0.940, 0.955, 0.961, 0.965) and MSE fell (0.00617→0.00355), confirming that a wider convolutional bottleneck retains more spatial detail, though gains shrink beyond 16 channels.

Exercise 2 – Gaussian vs. salt-and-pepper denoising (15 epochs, same architecture): Gaussian noise ($\sigma = 0.2$) gave MSE 0.00469/SSIM 0.946; salt-and-pepper ($p = 0.10$) gave MSE 0.00513/SSIM 0.944 – comparable difficulty overall, with salt-and-pepper very slightly harder at this configuration.

Exercise 3 – Denoising at two noise levels (15 epochs): SSIM was 0.962 at $\sigma = 0.1$ versus 0.917 at $\sigma = 0.3$ when each model was trained and tested at its own noise level, confirming that higher corruption consistently lowers achievable reconstruction quality.

Exercise 4 – Conv2DTranspose vs. UpSampling2D decoder (20 epochs): Conv2DTranspose slightly outperformed UpSampling2D (SSIM 0.978 vs. 0.975, MSE 0.00209 vs. 0.00242) at the cost of more parameters (111,425 vs. 74,497), since learned transposed-convolution kernels can recover detail that fixed upsampling cannot.

Exercise 5 – VAE latent dim 2 vs. 8 (20 epochs): increasing the latent dimension from 2 to 8 substantially improved reconstruction (SSIM 0.517→0.778, MSE 0.0445→0.0208), showing that 2-D latent capacity is the main bottleneck limiting VAE reconstruction quality.

Exercise 6 – Effect of KL weight β (20 epochs): at $\beta = 0.5, 1.0, 5.0$ the KL loss fell (6.57, 5.60, 3.43) while reconstruction loss rose slightly and SSIM dropped marginally (0.494, 0.496, 0.469) – a larger β compresses the latent space more aggressively toward the prior, trading reconstruction fidelity for a more regularized latent space.

Exercise 7 – Larger 10×10 generated grid: the mean per-pixel standard deviation across 100 generated samples was 0.1160, confirming meaningful pixel-level diversity rather than the decoder collapsing to near-identical outputs.

Exercise 8 – Interpolation between digit 3 and digit 8 latent regions: the decoded sequence changed smoothly across all 11 interpolation steps (see `exercise8_interpolation.png`), reaffirming the latent space's local continuity for a second, independent digit pair.

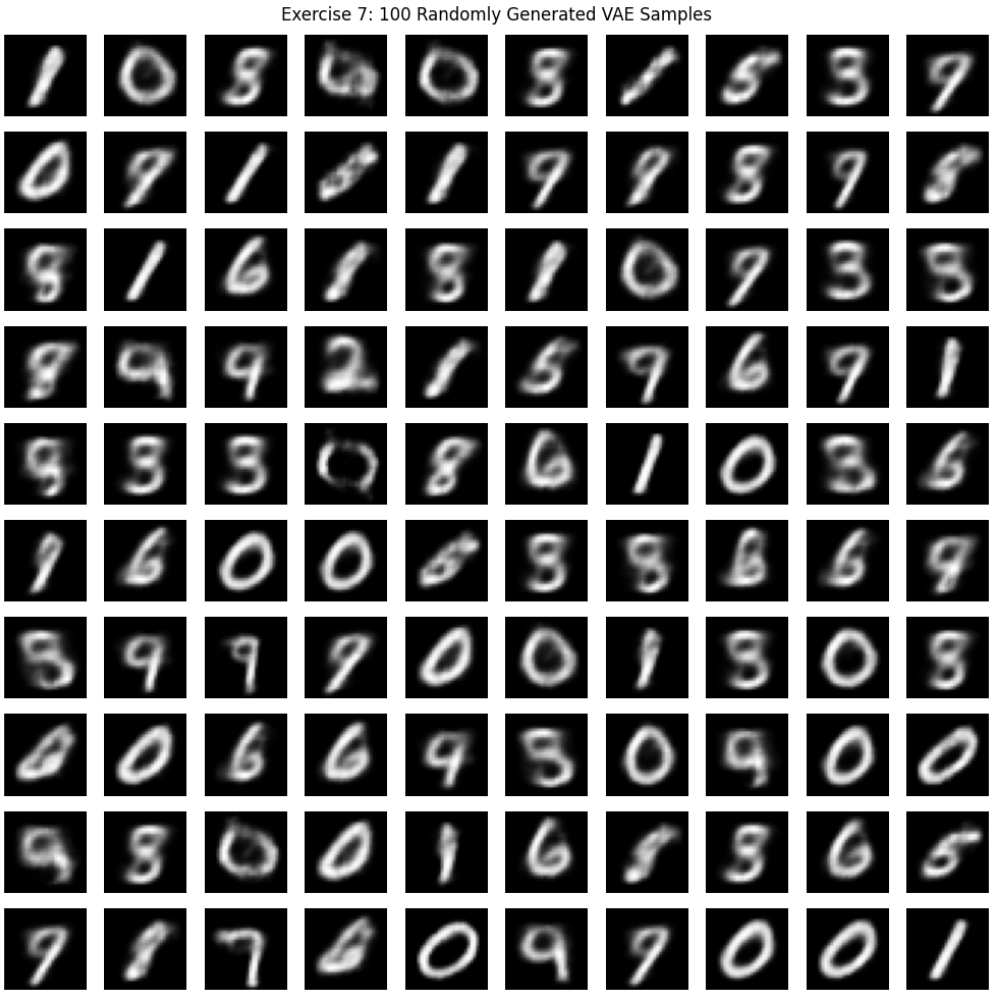


Figure 14: Exercise 7: 100 randomly generated VAE samples ($z \sim \mathcal{N}(0, I)$).

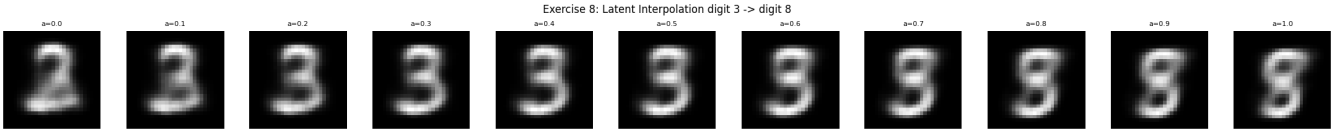


Figure 15: Exercise 8: latent-space interpolation between a digit-3 and a digit-8 test sample.

30 Expected Outcome

The completed pipeline demonstrates the full image-representation and reconstruction workflow (Image → Encoder → Latent Representation → Decoder → Reconstructed Image), the denoising workflow (Clean Image → Noise → Denoising Autoencoder → Recovered Image), and the variational generative workflow (Image → (μ, σ) → z → Decoder → Generated/Reconstructed Image), each supported by quantitative evaluation (MSE, MAE, SSIM) and visual analysis obtained from the student's own execution.

31 Conclusion

This experiment implemented and compared four autoencoder variants on a 10,000/2,000-image MNIST subset. The FC-AE (latent 16) reached test MSE 0.0223 / SSIM 0.746, while the CAE achieved substantially better reconstruction (MSE 0.0024, SSIM 0.975) with fewer parameters, confirming that preserving spatial structure through convolution is more effective than flattening. The denoising CAE, trained on Gaussian noise ($\sigma = 0.2$), reached SSIM 0.948 and degraded predictably as noise increased, recovering global digit structure even under stronger corruption. The VAE (2-D latent) reached reconstruction loss 151.6, KL loss 6.24, MSE 0.0445 and SSIM 0.517 – lower reconstruction fidelity than the deterministic models, but its regularized latent space produced a smoothly clustered visualization, recognizable generated digits, and continuous interpolation between digit pairs. The latent-dimension

study showed reconstruction quality improving monotonically with capacity with diminishing returns beyond dimension 16. Across all eight additional exercises, wider convolutional bottlenecks, transposed-convolution decoders, and larger VAE latent spaces consistently improved reconstruction quality, while increasing the KL weight traded reconstruction sharpness for stronger latent regularization. Overall, the results confirm that deterministic convolutional autoencoders are best suited for pure reconstruction, while VAEs trade some reconstruction accuracy for a structured, generative latent space.

32 Source Code

The complete implementation was carried out in Python using Google Colab.

`https://github.com/K-Mithran/Deep-Learning/Lab7/DL_7.ipynb`

The complete source code used in this experiment is available in the submitted Google Colab notebook and GitHub repository.

33 References

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